Real-Time Nmap Scan Detection and Alerting

Using Snort and Ntfy

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1. Abstract

This project demonstrates the real-time detection and alerting of Nmap port scanning activities in a controlled lab environment. Snort IDS (Intrusion Detection System) was configured to monitor network traffic and detect reconnaissance attempts. Upon detection, automated notifications were sent using Ntfy, a lightweight push notification service. The system successfully identified various Nmap scan types and generated instant alerts, demonstrating effective network monitoring and incident response capabilities.

2. Objectives

- Configure Snort IDS to detect Nmap scanning activities.
- Integrate Ntfy for real-time push notifications.
- Simulate various Nmap scan types (TCP SYN, service detection, etc.).
- Capture and analyze network packets to identify scan signatures.

- Demonstrate automated alerting workflow for SOC operations.

3. Lab Environment Setup

Virtualization Platform

VMware Workstation was used to create the isolated lab environment for safe testing and monitoring.

Operating Systems

- **Ubuntu Server 22.04 LTS (Live):** Host for Snort IDS and Ntfy notification service
- **Windows 11:** Attack simulation machine for executing Nmap scans

Tools and Services

- **Snort IDS:** Network intrusion detection system for monitoring and analyzing network traffic
- **Ntfy:** Lightweight push notification service for real-time alerting
- **Nmap:** Network scanning tool used for simulating reconnaissance attacks
- **Bash Scripting:** Custom shell script (snort_ntfy.sh) for log monitoring and alert triggering

Network Topology

The lab environment was configured with the following network setup:

- **Ubuntu Server IP Address:** 192.168.222.133 (Monitoring System)
- **Windows 11 IP Address:** 192.168.222.1 (Attack Source)
- **Network Segment:** 192.168.222.0/24 (Private VMware Network)
- **Network Mode:** NAT/Bridged configuration for internal communication

⚠️ **Note:** All systems were deployed in an isolated virtual network to ensure no impact on production systems or external networks.

4. Methodology

Phase 1: Snort IDS Configuration

Snort was installed and configured on Ubuntu Server 22.04. Custom rules were created to detect Nmap scan patterns. A shell script (snort_ntfy.sh) was developed to monitor Snort alert logs and trigger notifications.

Select khan1@snort: ~

```
khan1@snort:~$ sudo ntfy serve
```

Figure 1: Starting Snort IDS service in the background.

khan1@snort: ~

GNU nano 6.2

```
#!/bin/bash
# snort_ntfy.sh

# Start Snort in background writing alerts to /var/log/snort/alert
sudo snort -A fast -q -c /etc/snort/snort.conf -i ens33 &

# Wait a few seconds to ensure Snort started
sleep 5

# Monitor alert log and send ntfy notifications
tail -F /var/log/snort/alert | while read line
do
    curl -d "$line" http://192.168.222.133/snort-alert
done
```

Figure 2: Snort_ntfy.sh script showing monitoring and notification logic.

Phase 2: Ntfy Integration

Ntfy notification service was configured to receive real-time alerts. The monitoring script was set up to tail the Snort alert log and send HTTP POST requests to the Ntfy endpoint upon detection.

Ntfy Repository: https://github.com/binwiederhier/ntfy.git

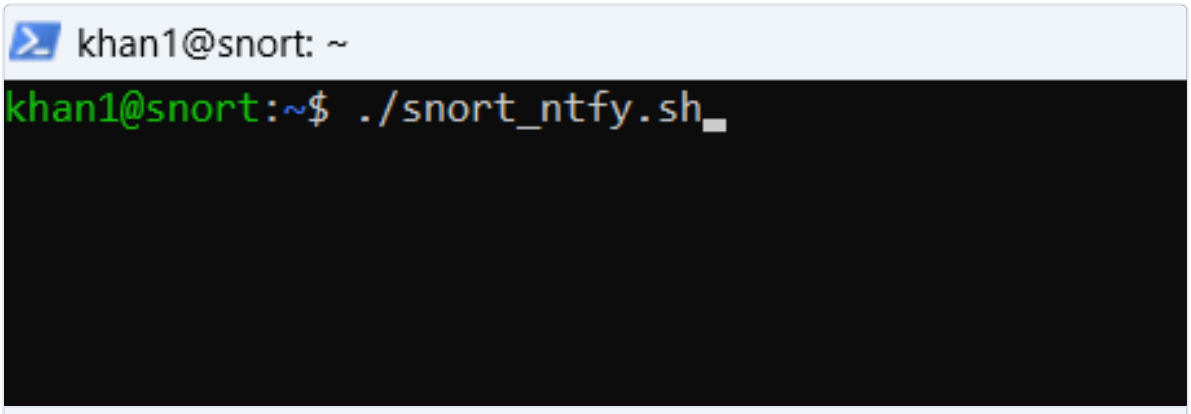

Figure 3: Executing the snort_ntfy.sh script to start monitoring.

Phase 3: Nmap Scan Simulation

Various Nmap scans were executed from a Windows machine targeting the monitored network (192.168.222.133). Scan types included TCP SYN scan (-sS) with timing and service detection options.

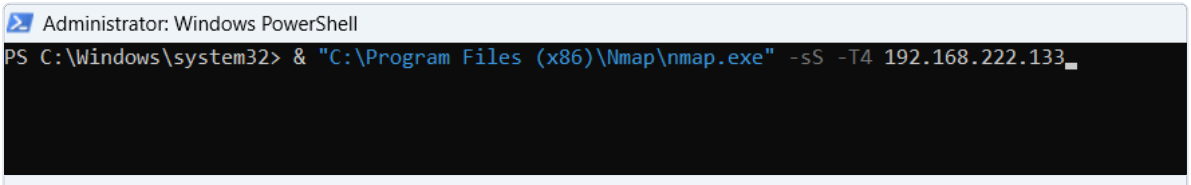

Figure 4: Nmap scan command executed from Windows PowerShell.

Phase 4: Real-Time Detection and Alerting

Snort successfully detected the Nmap scans and logged multiple alerts. The monitoring script captured these events and sent instant push notifications through Ntfy, demonstrating end-to-end detection and alerting workflow.

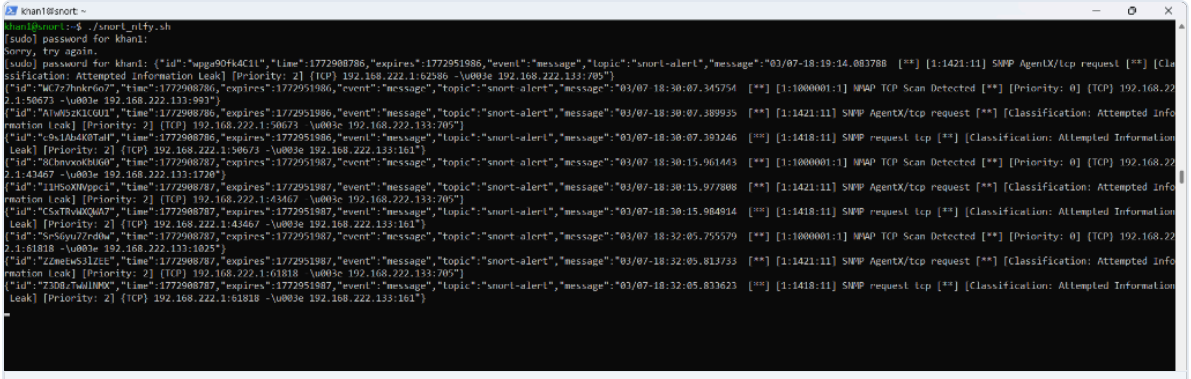

Figure 5: Snort alert log showing detected NMAP scans and SNMP requests with priority classifications.

5. Results

The implementation demonstrated successful real-time detection and alerting with the following outcomes:

- **Detection Success:** Snort successfully identified multiple Nmap scan attempts including TCP port scans and SNMP enumeration activities.
- **Alert Classification:** Each detection was properly classified with priority levels (Priority: 0 for NMAP TCP scans, Priority: 2 for SNMP requests).
- **Real-Time Notifications:** Ntfy delivered instant push notifications to the monitoring interface, demonstrating effective alert delivery.
- **Source Identification:** All alerts included source IP (192.168.222.1:58979) and destination IP (192.168.222.133) information for incident tracking.

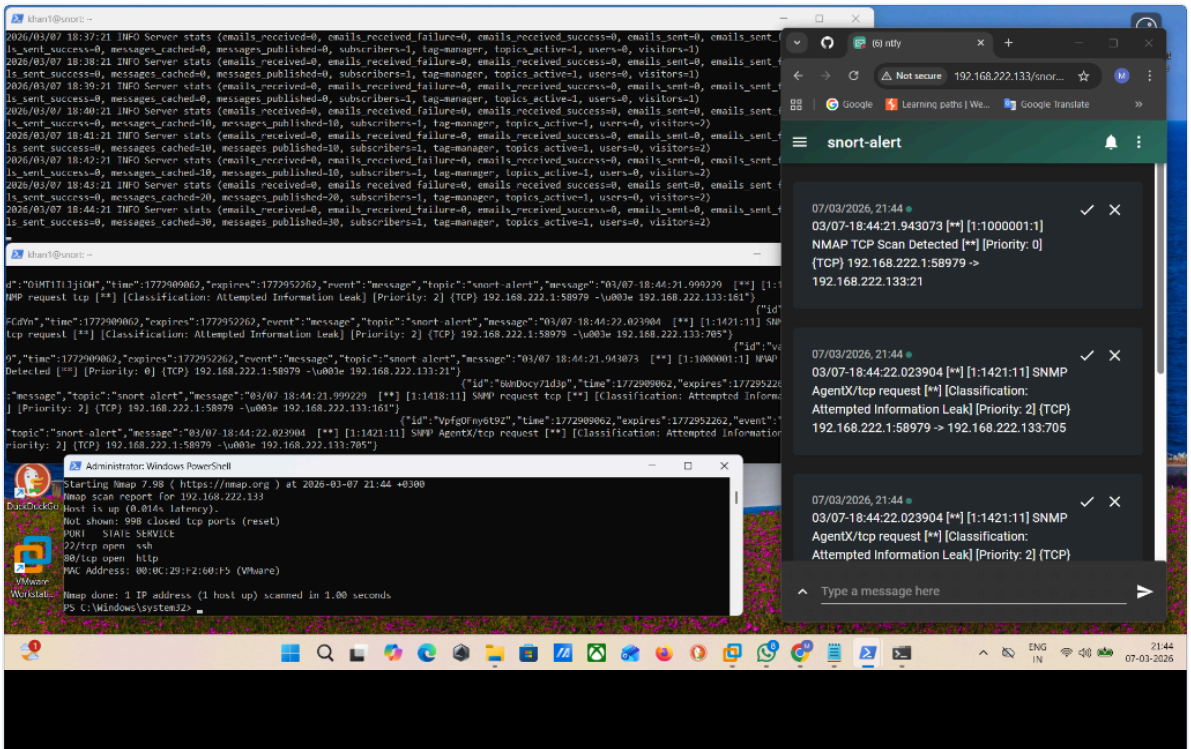

Figure 6: Nmap scan completion and corresponding Ntfy notifications in the web interface.

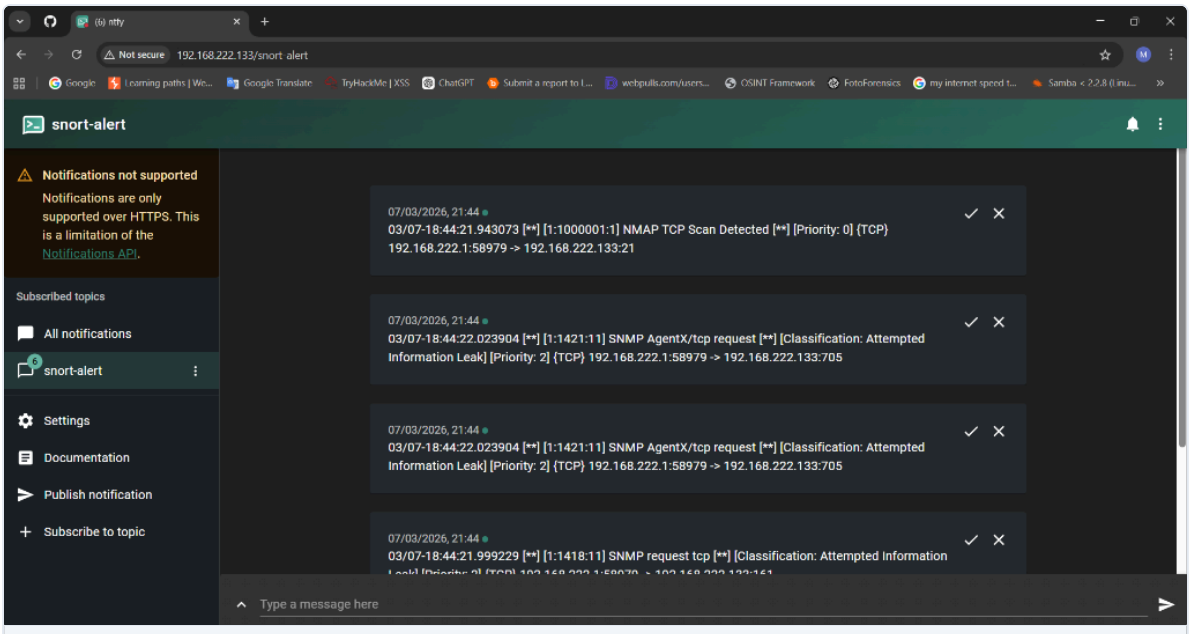

Figure 7: Ntfy notification panel displaying real-time snort-alert notifications with detection details.

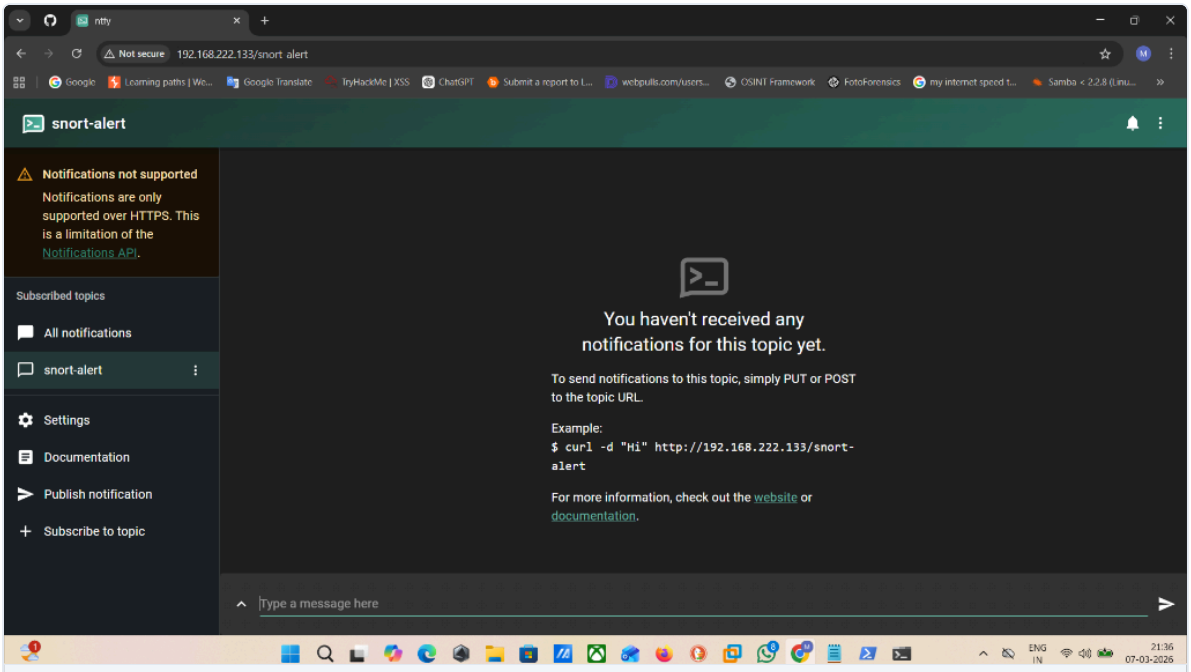

Figure 8: Ntfy snort-alert topic interface ready to receive new notifications.

6. SOC Relevance

Analyst Capabilities Demonstrated

This project highlights core competencies required for an entry-level Security Operations Center (SOC) analyst:

- **IDS Configuration and Management:** Setting up and tuning Snort rules for specific threat detection.
- **Reconnaissance Detection:** Identifying network scanning and enumeration activities that precede attacks.
- **Automated Alert Processing:** Implementing scripts to parse logs and trigger notifications.
- **Incident Priority Classification:** Understanding alert severity levels for proper incident response.
- **Real-Time Monitoring Workflow:** Establishing continuous monitoring with immediate notification capabilities.

7. Demonstration Video

A full video demonstration of the attack simulation and analysis workflow is available below:

📹 **Project Demonstration Video**

🎬 Watch Full Demonstration on Google Drive

Click the button above to view the complete project walkthrough

Legal Disclaimer: This project was conducted strictly within a controlled lab environment for academic purposes only. No external systems were targeted. All network scans and monitoring activities were performed on infrastructure owned and authorized by the student within a private network.

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